Literature Review

Yang and Wang (2025), in their study titled *“Random Forest-Based Machine Failure Prediction: A Performance Comparison,”* investigated the effectiveness of machine learning algorithms for predictive maintenance in industrial systems. The study aimed to identify reliable models for predicting machine failures by comparing traditional machine learning algorithms, ensemble methods, and deep learning approaches.

The researchers evaluated twelve machine learning models using two publicly available datasets: the Machine Predictive Maintenance Classification dataset, containing approximately 10,000 records, and a machine failure prediction dataset containing 943 records. The study addressed important challenges in predictive maintenance, including class imbalance, feature selection, model generalization, and the interpretation of model predictions. Data preprocessing techniques, hyperparameter tuning, and methods for handling imbalanced classes were used to improve model performance.

The models were evaluated using several performance metrics, including accuracy, precision, recall, F1-score, and ROC AUC. According to the reported results, Random Forest achieved an accuracy of 99.50% on the first dataset and 88.73% on the second dataset. These findings suggest that Random Forest can effectively identify patterns in industrial sensor data and provide promising results for machine failure prediction. The study also examined feature importance to identify influential variables, such as torque, tool wear, rotational speed, and sensor readings.

Despite these promising results, the study acknowledged several limitations. The evaluation was based on only two datasets, one of which was synthetic, and further validation using real-world industrial data was considered necessary. In addition, the study focused primarily on binary failure prediction, leaving opportunities for future research into predicting specific failure types and integrating models with real-time industrial monitoring systems.

Overall, this study provides a useful foundation for predictive maintenance projects by demonstrating the potential of Random Forest and highlighting the importance of comprehensive model evaluation. It supports the investigation of Random Forest as a candidate model for the proposed project, while emphasizing the need to validate its performance on the selected dataset rather than assuming that the same accuracy will be achieved.

Comparative Analysis of Machine Learning Algorithms for Predictive Maintenance

This study compares five machine learning algorithms for industrial equipment failure prediction using the AI4I 2020 Predictive Maintenance Dataset. The evaluated models include Random Forest, Decision Tree, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Logistic Regression.

The study applies data preprocessing, an 80/20 train-test split, SMOTE to address class imbalance in the training data, and Grid Search for hyperparameter optimization. Model performance is evaluated using accuracy, precision, recall, and F1-score.

According to the reported results, Random Forest achieved the best overall performance, with an F1-score of 97.10% and a recall of 98.53%. The findings emphasize that accuracy alone may be misleading when equipment failures are rare.

However, the study acknowledges that the dataset is synthetic and that the evaluated models do not explicitly capture temporal degradation patterns. The authors suggest investigating sequential models such as LSTM and GRU, along with real industrial datasets, in future work.

Relevance to our project: This study provides a useful baseline for selecting and evaluating classification models. It also highlights the importance of handling class imbalance, preventing data leakage, and evaluating failure detection using recall and F1-score. Our project can build on these findings by exploring temporal features and

developing a deployment and monitoring pipeline, if these components are supported by our available data and project scope.

Suggested Improvements

To further enhance the performance, scalability, and practical applicability of the AI-Powered Predictive Maintenance System in real-world industrial environments, the following improvements are proposed:

1. Integration with Edge Computing Devices

Proposal: Deploy the trained model lightweight version (e.g., ONNX, TensorFlow Lite) directly on Edge Devices such as Raspberry Pi or NVIDIA Jetson Nano situated near industrial machinery.

Impact: Enables real-time, low-latency failure inference directly at the machine level without depending on continuous high-bandwidth internet connections to the cloud.

2. Model Explainability and Interpretability (Explainable AI – XAI)

Proposal: Incorporate Explainable AI frameworks such as SHAP (Shapley Additive exPlanations) or LIME.

Impact: Provides maintenance engineers with clear insights into why the model predicts a failure (e.g., highlighting that a 15% increase in vibration combined with elevated temp is the primary root cause), building trust in automated predictions.

3. Automated Retraining & Advanced MLOps Pipelines

Proposal: Upgrade the current MLOps deployment setup by creating fully automated retraining pipelines triggered by Model Drift or Data Drift detection algorithms.

Impact: Ensures the system maintains high accuracy over time as industrial equipment undergoes natural wear and operational conditions change.

4. Multi-Modal Sensor Data Fusion

Proposal: Expand data ingestion beyond numerical IoT sensor streams (temperature, vibration, pressure) to include acoustic emission sensors and thermal infrared imaging.

Impact: Combines visual and auditory degradation patterns to capture micro-faults at an even earlier stage before physical structural wear manifests.

5. Generative AI for Anomaly & Synthetic Failure Data

Proposal: Implement Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs) to generate realistic synthetic failure scenarios.

Impact: Overcomes the severe class imbalance inherent in real-world industrial datasets where actual failure events are extremely rare.

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